Joy Zhang

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EDUCATION

Cornell University, College of Engineering
Master of Engineering, Computer Science, GPA 3.9 **May 2021**
Bachelor of Science, Computer Science Major, Game Design Minor, GPA 3.5 **Dec 2020**

PROFESSIONAL EXPERIENCE

Sandtable, *Game Engineer* **May 2025 – Apr 2026**

- Developed a cross-platform (PC, tablet, web, mobile) multiplayer strategy game in **Unity** by designing game systems and implementing the core gameplay loop, game mechanics, and networked gameplay
- Implemented hierarchical finite state machines to manage the game state
- Created flexible UI and theme system to streamline development and built UI from low-fidelity designs
- Wrote shaders and added in-game visuals and effects such as blur to the render pipeline
- Used real-world data and maps to set the game anywhere in the world, performed complex geospatial calculations, and explored standard geospatial libraries such as PROJ, GDAL, GeoJSON, and EPSG

Hometopia Inc.
Senior Game Developer **Dec 2024 - Feb 2025**
Junior Game Developer **Jul 2024 - Dec 2024**

- Developed new features and wrote clean, well-organized, and easily maintainable **C#** code for an online multiplayer house-building PC sandbox game made in **Unity**
- Owned UI development and collaborated with designers to give the game a cohesive look and feel
- Led and mentored junior developers, taught Unity UI, delegated tasks, and performed code reviews
- Developed an audio system to centralize and streamline control of music, ambience, and sound effects
- Improved player onboarding by creating a system for the first-time user experience and tutorial
- Improved backend multiplayer features, including friend list, parties, chat, and save files
- Investigated and fixed critical gameplay, multiplayer, and UI bugs using debugging tools

Northrop Grumman
ODIN, *Game Developer* **Jan 2022 - Jun 2024**

- Developed Omniverse apps in **Python** with Universal Scene Description to build a virtual warfighting environment by controlling drones and sensors, streamlining data generation, and visualizing data
- Built physically accurate flight simulators for vertical take-off and landing drones with Omniverse's integrated **PhysX** physics engine and performed manual 3D vector math calculations
- Created a data generation pipeline for Omniverse to train AI/ML models by developing a suite of tools in **Python** to create, control, and randomize scenes, assets, and perceptual variables

COAST, *Software Engineer* **May 2023 – Jun 2024**

- Fixed bugs and implemented updates in **C++** in Linux to maintain system health

COAGEN, *Software Engineer* **Aug 2021 - Dec 2021**

- Built demo in **Java** for a long short-term memory (LSTM) predicting adhoc interruptions to schedule

PROJECTS

Game Design Initiative at Cornell **Jan 2019 – May 2021**

- Developed gameplay mechanics, physics, and programmatic animations in **C++** and in **Java** with **libGDX** and **Box2D** as a gameplay programmer to make games in cross-functional teams of 7
- Accepted to BostonFIG Fest with computer game *Starstruck* (2019) and mobile game *Spectacle!* (2020)

Cornell Big Red Hacks **Sep 2019**

- Awarded Best Design Hack for *One Call Away*, a short story platformer created in **Unity** in 36 hours

SKILLS

Competencies: Agile, Algorithms, Data structures, Game development, Gameplay programming, Object-oriented programming, Physics simulation, UI programming
Programming Languages: C, C#, C++, Java, MATLAB, OCaml, Perl, PHP, Python
Specialties: Blender, Box2D, Construct3, libGDX, Nvidia Omniverse, PhysX, Pygame, Unity
Other Software: ClickUp, CSS, Docker, Eclipse, Figma, Git, HTML, IntelliJ IDEA, Jira, LaTeX, Linux, Maven, Robot Framework, Unity Version Control, Universal Scene Description, Visual Studio, Visual Studio Code